

Nios[®] V/m Processor PIO Design

Agilex[™] 5 FPGA

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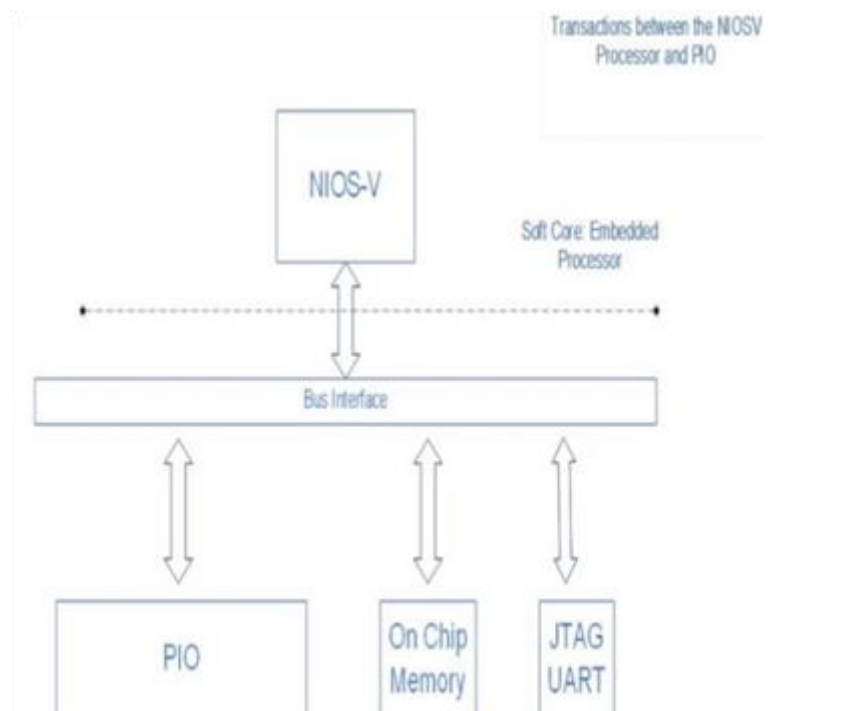
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1. Theory of Operation

This design demonstrates the transaction between the Nios® V processor and the Parallel Input/Output (PIO) core for the Agilex™ 5 FPGA E-Series 065B Premium Development Kit (ES1) DK-A5E065BB32AES1.

The PIO core is configured for output ports only and the outputs are connected to the LED on the development kit. The application, which runs atop this design, toggles these output registers of the PIO core. The application writes and reads back the content from the IP location.

Figure 1-1. Block Diagram



1.1. IP Cores

The following IPs are used in this design.

- Nios V/m soft processor core
- On Chip RAM-II
- JTAG UART
- PIO IP
- System ID

2. Directory Structure

The directory structure is explained below:

Root directory: `niosv_m/niosv_m_pio`

Directories	Content
docs	Document capturing the details of the example design
img	Block diagram for the example designs
ready_to_test	Prebuilt binaries (FPGA <code>.sof</code> file and Application <code>.elf</code> file) which can be programmed on the board and tested
sources - hw - scripts - sw/app	Files needed to create the design - Necessary hardware files (<code>.sv</code>, <code>.v</code>, <code>.ip</code>) of the design - This folder consists of scripts to build the design - This folder contains software application files

3. Quick Start Guide

You may try the example on the target development kit using the prebuilt binaries. The respective FPGA `sof` and Application `elf` files are found in `ready_to_test` folder.

Refer to [Chapter 6 Programming the Design](#) for the steps to program these files.

4. Generating the Design

The implementation of Nios® V processor system requires a hardware design developed using the Quartus® Prime software. After that, it is required to develop the software design that complements the processor system.

4.1. Building Hardware Design

Begin designing the system hardware by instantiating the Nios® V processor and its peripherals into Platform Designer. After configuring the system assignments and constraints, complete the hardware design by performing a successful compilation.

We will be using `sources/scripts/build_sof.py` to develop the hardware design.

This script will create a new platform design, add the IPs into the design, makes the connections, compiles the design and generates the `.sof` file.

Steps to execute the script are as follows:

1. Invoke the Nios V Command Shell in the terminal
2. Run the following command in the terminal from top level project directory:
`> quartus_py ./scripts/build_sof.py`
3. The Quartus tool will compile the design and generate the output files into `<Root Directory>/sources/hw/output_files`

4.2. Building Software Design

After the processor system is ready, you may begin building the software design. It consists of the following steps:

1. Create a board support package (BSP) project.
2. Create a Nios® V processor application project with source code.
3. Build the application.
4. Convert the application `.elf` file into `.hex` file for memory initialization.

To create software application, run the following commands in the terminal:

```
> niosv-bsp -c --quartus-project=hw/top.qpf --qsys=hw/qsys_top.qsys --  
type=hal sw/bsp/settings.bsp  
> niosv-app --bsp-dir=sw/bsp --app-dir=sw/app --srcs=sw/app/main.c  
> niosv-shell  
> cmake -S ./sw/app -B sw/app/build -G "Unix Makefiles"  
> make -C sw/app/build  
> elf2hex sw/app/build/app.elf -b 0x0 -w 32 -e 0x9ffff  
sw/app/build/onchip_mem.hex -r4
```

Note: It is recommended to clean the `app/build` project before regenerating `elf` (`cmake` and `make`).

5. Simulating the Design

You may carry on simulation for this design to evaluate its intended functions. During simulation, the action of downloading .elf file is unable to be simulated. Thus, the processor memory is instead initialized with the application .hex file.

Use the following commands to run the simulation:

```
# Generate Testbench from Platform Designer.
# Generate -> Generate Testbench System

> cp ./sw/app/build/onchip_mem.hex ./qsys_top_tb/qsys_top_tb/sim/mentor
> cd hw/qsys_top_tb/qsys_top_tb/sim/mentor/
> vsim &
> source msim_setup.tcl
> ld_debug
> run -all
```

6. Programming the Design

Use the Quartus® Prime Programmer tool and the Ashling* RiscFree* IDE to program the Nios® V processor-based system into the FPGA and to run your application.

Program the generated sof, then download the elf file on the board according to the command below.

```
> quartus_pgm --cable=1 -m jtag -o 'p;top.sof'

#----- Download the elf file on the board ----#
> niosv-download -g ready_to_test/app.elf -c 1

#----- Verify the output on the terminal by using the following command in the
terminal ----#
> juart-terminal -d 1 -c 1 -i 0
```

Note: Reduce the JTAG clock frequency to 6MHz before programming the application .elf file on the board using the command: `jtagconfig --setparam 1 JtagClock 6M`

7. Expected Results

The following is the output as observed on the JTAG UART terminal. The output is analogous to the logic from the application code. Users should be able to observe the same output on their terminal/setup.

Figure 1-2. Expected Result

```
# Application to toggle the PIOs- [4:0]
# DATA READBACK FROM PIO_0_BASE is 0x0
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x1
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x2
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x3
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x4
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x5
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x6
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x7
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x8
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x9
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xa
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xb
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xc
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xd
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xe
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xf
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x0
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x1
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x2
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x3
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x4
```

```
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x5
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x6
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x7
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x8
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x9
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xa
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xb
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xc
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xd
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xe
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xf
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x0
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x1
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x2
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x3
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x4
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x5
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x6
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x7
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x8
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x9
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xa
```



```
# DATA READBACK FROM PIO_0_BASE is 0xb
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xc
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xd
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xe
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xf
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x0
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x1
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x2
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x3
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x4
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x5
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x6
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x7
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x8
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x9
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xa
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xb
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xc
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xd
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xe
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xf
# DATA MATCHED - TEST PASSED
# NIOSV-PIO Test PASSED
```